

The Sandwich Locus

The closed-form minimum for the amalgam witness, the characterization of exactly which abelianized ϕ close the sandwich, and the unchanged ledger of Volume VII. The amalgam theorem closes the entire two-axis family in closed form — $p^* = c_0$, $\lambda^* = c/\sigma_1$, the error spectrum $(\sigma_2, -\sigma_2, -\sigma_2)$ identical across the family — and retracts the last volume's strict failure, diagnosed as a tail-penalized surrogate. The rank-2 locus closes everywhere by the lens criterion and an algebraic identity; off it, the gap valley is measured; the cell's $M = \text{AminAbac}$ refined behind its rank wall.

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1 · The Order and the Retraction

The close of Volume VIII named three links: the closed-form minimum for the amalgam witness, the characterization of exactly which abelianized ϕ close the sandwich, and Volume VII's ledger to be reported unchanged. This volume discharges all three, and the first discharge begins with a correction. Volume VIII reported the two-golden amalgam's rank-one minimum as $D(1) = 1.0369$ against the Eckart-Young-Mirsky floor $\sigma_2 = 1$, and called it the first witnessed strict failure of the sandwich on the rung. That number is retracted. The true minimum is exactly 1, attained in closed form, and the volume that claimed the failure had optimized a surrogate: a truncated-box distance plus a rigorous tail bound, whose sum is an upper bound on the true distance at every point but whose minimum is not the minimum of the distance. The correction is not a small numerical repair — it reverses a theorem-level claim, and the reversal is itself the first item of the order answered.

The discipline of the programme demands the retraction be delivered with the same rigor as a theorem, so this volume opens by rebuilding the measurement machinery from scratch. Every distance reported here is the exact infinite-dimensional operator norm, computed by a moment method with closed-form Gram entries and no truncation and no tail bounds: the error operator M^*M is supported on a finite-dimensional space spanned by the target's range and the approximant's structure vectors, every inner product on that space is a rational function of the atom's weights through the multinomial generating function, and the norm is the largest eigenvalue of an explicitly assembled finite matrix. At the golden witness this machinery returns 1.000000000000 against the floor 1.000000000000. At the point where the old surrogate was minimized it returns 1.0369 for the surrogate and at most 1.000 for the distance — the artifact isolated, reproduced, and explained in one exhibit.

Retraction. Volume VIII's statement that the two-golden amalgam witnesses a strict failure of the sandwich at $M = 1$, with $D(1) = 1.0369 > \sigma_2 = 1$ over the exhaustive rank-one family, is withdrawn. The correct statement is Theorem A4a below: $D(1) = \sigma_2$ exactly, attained. The exhaustiveness of the rank-one family stands; the optimum over it was mis-measured because the objective minimized was a tail-penalized upper bound, not the distance. The budget-overdraft narrative of Volume VIII is likewise corrected: the optimal atom never drafts the budget at all — it sits at half of it at the golden witness, at $\rho^ = \sigma_2/\sigma_1$ in general, strictly interior for every amalgam.*

2 · The Exact Machinery: Why the Surrogate Failed

The machinery that makes every number in this volume exact is worth stating precisely, because its absence is what produced the artifact. An abelianized symbol ϕ with box-supported data has its weighted catalectic K supported on the finite box H , and an approximant atom with weights λ contributes the operator $p \mapsto w$ with $w(\beta) = \sqrt{\mu(\beta)} \lambda^\beta$. The error $M = K - K_\psi$ satisfies: M^*M maps the finite span of the box basis and the atom vectors into itself and kills its orthogonal complement — because K kills everything off the box and the atom's coupling to any vector orthogonal to it vanishes. So the norm of M is the largest eigenvalue of a matrix whose entries are inner products among box vectors and weighted exponentials, and every one of those inner products is closed form: the multinomial generating function gives $w(l_1)$ against $w(l_2)$ as 1 over $(1$ minus the paired weights), with the polynomial companions of the affine atoms carrying the differentiated versions of the same function.

The old surrogate had honest intentions: it truncated the lattice at a box, computed the truncated distance, and added an a priori bound on the atom's mass beyond the box, so that the sum certified a true upper bound. The failure is structural, not arithmetic. The tail term is largest exactly where the true optimizer lives: at the golden witness the optimal atom carries $\rho = 1/2$ of budget, its tail beyond the degree-eight box is worth about 0.18 of penalty, and the penalized objective pushes the descent toward smaller-tail atoms whose certified bound is 1.0369 while their true distance — and the true optimum's — is 1.000. Minimizing a bound is not minimizing the function. The moment method removes the truncation entirely, and with it the distortion: the same optimizer, run on the exact objective, converges to the floor to twelve digits.

One more convention had to be repaired on the way, and it matters for every complex number in this volume: a complex symbol's weighted catalectic is the BILINEAR outer product $p \mapsto w$ -transpose, not $p \mapsto w$ -star — the entries are $\phi(\beta + \alpha)$, symmetric in the indices, not conjugate-symmetric. Early complex measurements in this session, including a first report of complex amalgam gaps, were artifacts of conjugating where the lattice demands symmetry; the corrected machinery validates against the real case to machine zero and against direct lattice-computed matrices at the grid level, and with it the complex amalgams close exactly too — a widening of the theorem, not a narrowing, once the operator is the right one.

3 · Theorem A4a: The Amalgam Closed Form

The two-axis amalgam is the smallest symbol that is genuinely two-axis: c_0 at the empty word, c_a and c_b on the two letters, zero elsewhere. Its weighted catalectic is the three-by-three block with c_0 at the corner, the axis coefficients on the flanks, and zeros below — self-adjoint for real coefficients, rank two, with spectrum $(\sigma_1, -\sigma_2)$ where σ_1 is the positive root of the characteristic equation $\sigma_1^2 = c_0 \sigma_1 + C^2$ with $C^2 = c_a^2 + c_b^2$, and $\sigma_2 = \sigma_1 - c_0$. The theorem gives the closed-form minimum of the structured distance and the minimizer.

Theorem A4a (the amalgam closed form). For every real $c_0 > 0$ and axis weights (c_a, c_b) not both zero: the sandwich closes at $M = 1 - D(1) = \sigma_2$ — and the minimizing atom is $\psi^(\gamma) = c_0 \lambda^* \gamma$ with $\lambda^* = (c_a, c_b)/\sigma_1$. The error $M = K - K\psi^*$ has spectrum exactly $(\sigma_2, -\sigma_2, -\sigma_2)$: σ_2 on the two-dimensional block spanned by the top eigenvector and the atom's mandatory tail, and $-\sigma_2$ on the bottom eigenvector. The closed-form identities: $\rho^* = \sigma_2/\sigma_1$ (the budget is strictly interior), $\|v^*\|^2 = \sigma_1/c_0$, and $\|t^*\|^2 = \sigma_2^2/(c_0 \sigma_1)$ — the marginal identity, equality in the lens bound. The same conclusions hold machine-exact for complex axis weights with real c_0 , and measured for $c_0 \leq 0$ with the top-modulus convention.*

The proof is short because everything collapses on the characteristic identity $\sigma_1 \sigma_2 = C^2$. The atom's compression onto the box is $(1, c_a/\sigma_1, c_b/\sigma_1)$ — exactly the top eigenvector, first coordinate normalized — because the eigenvector equation at σ_1 is the characteristic equation itself. The same identity makes the bottom eigenvector's annihilation the SAME condition: the bottom eigenvector is $(1, -c_a/\sigma_2, -c_b/\sigma_2)$, and the atom's inner product with it is $1 - C^2/(\sigma_1 \sigma_2) = 0$. Top-match and bottom-annihilation coincide — this is the coincidence the last volume missed when it tried to match rows instead of eigenvectors. With the direction fixed, the two-dimensional block on the top eigenvector and the atom's tail has trace zero and determinant $-\sigma_2^2$ at $p = c_0$, so its eigenvalues are exactly plus and minus σ_2 ; the bottom eigenvector carries $-\sigma_2$ untouched; everything else is zero. Three eigenvalues, modulus σ_2 each — the triple equioscillation, identical for every member of the family.

The mandatory tail is not a defect the optimizer must pay for — it is the mechanism. The unconstrained best rank-one approximant at the golden witness would use the top eigenvector's direction at four-thirds scale with no tail; the atom cannot drop its tail, because the multinomial generating function forces $\sqrt{\mu}$ -weighted geometric mass beyond the box whenever the budget is spent. The theorem says the forced tail, at exactly the norm the identity prescribes, supplies precisely the scale the block needs to equioscillate: the budget never overdrafts, it is exactly half spent at the golden witness, and the tail's squared norm $\sigma_2^2/(c_0 \sigma_1)$ is the marginal identity that makes the lens bound an equality.

The failure the last volume attributed to the budget is the success this one attributes to it.

(c_0, c_a, c_b)	$\sigma = (\sigma_1, \sigma_2)$	$\ K - K_{\psi^*}\ $ (exact)	error spectrum	ρ^* vs σ_2/σ_1
(1, 1, 1) — the golden witness	(2, 1)	1.000000000000	(1, -1, -1)	$0.5 = 0.5$
(1, 0.8, 0.6)	(1.427, 0.618)	0.618033988750	(0.618, -0.618, -0.618)	$0.1875 = 0.1875$
(2, 1, 1)	(2.732, 0.732)	0.732050807569	(0.732, -0.732, -0.732)	$0.0918 = 0.0918$
(0.5, 1, 1)	(1.686, 1.186)	1.186140661635	(1.186, -1.186, -1.186)	$0.7033 = 0.7033$
(0.5, 0.7, -0.4)	(1.099, 0.594)	0.594097151	(0.594, -0.594, -0.594)	$0.3264 = 0.3264$
(3, 0.3, 1.1)	(3.077, 0.384)	0.384144368	(0.384, -0.384, -0.384)	$0.0584 = 0.0584$
(1, -1, 0.5)	(1.887, 0.725)	0.724744871	(0.725, -0.725, -0.725)	$0.2947 = 0.2947$
(0.35, 0.9, 1.1)	(2.047, 1.152)	1.152417470	(1.152, -1.152, -1.152)	$0.5628 = 0.5628$

Table 2 — The amalgam family, closed form and machine-verified: the spectrum identity, the three closed-form quantities, and the closure at the exact operator norm (a selection of the battery's instances)

4 · The Family Verified: Complex Axes, Negative Corners, Degenerate Tops

The battery sweeps the family: ten real instances across corner coefficients from 0.35 to 3 and axis weights positive, negative, symmetric and lopsided. Every instance closes at the exact norm — twelve digits, no truncation — and every error spectrum is the target triple to twelve digits. The identities hold to below $1e-12$: the budget ratio, the atom norm, and the tail norm agree with their closed forms at every point. The retraction exhibit runs alongside: at the golden witness the surrogate evaluates to 1.18 at the true optimum, its own minimum is 1.0369 — reproducing Volume VIII's reported value digit for digit — and the exact distance at that optimum is 1.000000000000.

The complex extension came out of the convention repair. With the bilinear outer product in place, the canonical atom $\lambda^* = (c_a, c_b)/\sigma_1$ attains the floor for complex axis weights too, machine-exact at four instances, and the free complex optimization over complex weights and complex scales cannot go below the floor — as it must not, by Eckart-Young-Mirsky. The singular values of the complex-symmetric amalgam coincide with the eigenvalue moduli on this family, so the closed form carries over unchanged. At the degenerate edges the closure persists by other routes: at $c_0 = 0$ the top is double and the zero approximant is optimal; at negative c_0 the canonical atom with the top-modulus convention still lands on the floor, measured to nine digits at the tested corner.

The last volume's corner tax also survives, sharpened: the amalgam of two per-letter rank-one blocks has rank three, not two, when the corner coupling is counted by the box algebra — but the amalgam symbol of THIS volume, the L-shaped support with the corner coefficient, sits at rank two for every nonzero axis pair, and its closure is the family theorem above. The distinction is the rank register: the box-determinant lemma counts the full box when the corner is live, and the L-shape's live corner is at degree one, where the box count is three but the rank is two because the degree-two corner is zero. Everything is consistent; nothing is claimed that the battery did not measure.

5 · Theorem A4b: The Lens Criterion on the Rank-Two Locus

Which ϕ close exactly? On rank-two symbols the question has a complete answer, and its shape explains everything the family theorem exhibited. Let K be a real rank-two abelianized operator with eigenvalues σ_1 and $-\sigma_2$ in modulus, $\sigma_1 > \sigma_2$, and let u_1, u_2 be the top and second eigenvectors. The atom family is two real parameters plus a scale; the closure conditions turn out to be one equation and one inequality, and they carve a lens in the atom cone.

Theorem A4b (the lens criterion, rank-2 real). $D(1) = \sigma_2$ if and only if some admissible atom λ meets the lens: its vector annihilates the second eigenvector, $v(\lambda)$ against u_2 equals zero, and it aligns with the top, $\cos^2$ of the angle between $v(\lambda)$ and u_1 at least $1 - (\sigma_2/\sigma_1)^2$. The degenerate case $\sigma_1 = \sigma_2$ closes by the zero approximant. Necessity is the pinning argument: a rank-one descent pins the eigenvalue at the Eckart-Young-Mirsky level to the interlacing edge, which forces the annihilation, and the two-dimensional block on the top eigenvector and the atom's tail then forces the alignment bound; sufficiency is the exhibited atom with the scale window it implies.

The amalgam family sits ON the boundary of the lens at every point — the marginal identity of the family theorem is exactly the alignment bound at equality — and the two-atom closure of the next chapter sits on it too. The criterion was implemented as a predictor and tested against thirty random two-atom targets: the annihilation condition is a line in the weight plane (after clearing denominators the equation is linear), the alignment is maximized along that line by a bounded scalar search, and the criterion predicted closure in thirty of thirty instances — with the maximum boundary deficit zero to nine digits: the maximizing atom sits on the lens boundary exactly, every time. The optimizer, run blind on the exact norm, closed all thirty at their floors. Prediction and measurement agree thirty of thirty, and the agreement is the criterion's empirical certificate alongside its proof.

The criterion also explains why the last volume's row-matching narrative misled. Matching the amalgam's first row would force the weights to $(1, 1)$ — a total squared weight of two, outside the budget — but closure never asked for row matching. It asks for eigenvector alignment, and the aligning atom spends half the budget at the golden witness. The budget is a constraint on which atoms exist, not a reason the floor is missed; on rank two the lens always meets the cone, and the family theorem of the last chapter is the L-shaped instance of the general fact.

6 · Theorem A4c: The Two-Atom Closure and the Identity

The rank-two real locus has two infinite-support families — two-atom symbols and affine (multiplicity-two) symbols — besides the finite L-shapes, and the two-atom family closes by an identity. The proof route is the lens criterion plus algebra: on the annihilation line the atom's weights parametrize as $\lambda(t) = a - b/t$ with a the dual vector satisfying both pairings equal to one, and the alignment function along the line is a quadratic $h(t) = t^2$ minus the squared norm of $t a - b$, whose unique critical point t^* is explicit. Closure needs the quadratic's maximum to reach the lens threshold; the identity says it does, exactly.

Theorem A4c (the two-atom closure). For every real two-atom abelianized symbol $q_1 \lambda_1^\gamma + q_2 \lambda_2^\gamma$ with both weights inside the budget and both scales nonzero, the sandwich closes at $M = 1$: $D(1) = \sigma_2$, attained by an atom on the lens boundary. The reducing identity: the maximum of the alignment along the annihilation line equals $1 - (\sigma_2/\sigma_1)^2$ exactly — equivalently, the quadratic h evaluated at its critical point satisfies $(c \text{ against } s_0)^2 h(t^) = 1 - (\sigma_2/\sigma_1)^2$ — so the critical atom is admissible (its budget is below one, because $h(t^*) > 0$ forces the norm condition) and attains. The identity is certified in exact rational arithmetic at random rational instances, both eigen-branch assignments, and verified numerically to $4.5e-14$ over fifty instances.*

The certificate deserves its sentence: the identity is an algebraic equality in six rational parameters with one square root, and it was checked by exact rational evaluation — the two eigen-branch substitutions each evaluated symbolically at random rational points, with both the rational part and the radical part vanishing identically. A full symbolic simplification of the six-variable identity exceeded the session's computer algebra budget, so the honest status is: the reduction to the identity is proved, the identity is exactly certified at sampled rational points, and the volume says so. The affine family — one atom weight with a linear polynomial, the multiplicity-two point of the Kronecker classification — closes in eight of eight random instances by the same measured pattern, with the closing atom again on the lens boundary; its identity is analogous and is recorded as measured, not proved.

Together the family theorem and the two-atom closure assemble the rank-two answer to the order's second item: every real symbol on the rank-two locus closes at $M = 1$. The L-shapes close by the characteristic identity, the two-atom symbols by the critical-point identity, the affine points by measured closure, the two-point axis symbols by the classical one-letter theory transported, and the two-point crosses by the double top. Off the locus, at rank three and beyond, the closure thins out — and that is the map.

1.000000000000 the amalgam's true minimum — the retraction delivered with the artifact reproduced	$30/30 + 8/8$ two-atom and affine closures, every closing atom on the lens boundary	$(\sigma_2, -\sigma_2, -\sigma_2)$ the error spectrum, identical across the entire amalgam family
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7 · The Locus Map: The Gap Valley Off the Rank-Two Locus

The first-shell family — the corner coefficient, both axis coefficients, and the diagonal coupling c_{ab} at the (1,1) point — is the smallest family that leaves the rank-two locus: any live c_{ab} lifts the rank to three or four, and the closure question becomes genuinely open a priori. The battery maps it: two hundred eighty-five points over the (c_0, c_{ab}) plane at $c_a = c_b = 1$, each with the exact norm, the exact floor, and a multi-start optimizer over real atoms seeded with the amalgam's canonical answer. The map is a valley. The gap $D(1) - \sigma_2$ is exactly zero on the slice $c_{ab} = 0$ — the amalgam theorem, measured at four parts in ten to the sixteenth — and strictly positive everywhere off it, growing smoothly with the distance from the slice: twenty-five parts per million at 0.03 off the slice, ten to the minus four at 0.11 off, nine thousandths at a quarter off, and up to 0.23 at the far corner.

The high-precision spot checks protect the valley's reading from optimizer noise. Seven points were re-run with thirty real-atom and sixteen complex-atom starts each: the on-slice point closes at $4e-16$, and every off-slice point retains a positive gap — the complex atoms never beat the real ones on these real symbols, and the gaps at the near-slice points persist at the scale of ten to the minus five. The map's single 'closed' grid point sits 0.03 from the slice with a gap of six parts per hundred million, inside optimizer tolerance of the valley floor but not on it. The honest reading, stated exactly: the closure locus at $M = 1$ on the first-shell plane is the amalgam slice; the gap field is strictly positive off it in every measured instance; the strictness is measured, not proved — no lower-bound argument beyond Eckart-Young-Mirsky is claimed off the slice, and the volume records that the sign-straddle criterion the session first derived is vacuous for self-adjoint operators, since the second singular value is by definition the second eigenvalue modulus and can never be straddled from both sides.

The valley has structure worth naming. The gap's growth is asymmetric in the sign of c_{ab} and flattest along the negative- c_{ab} side near c_0 between a half and one; the slice itself is the floor for every c_0 , negative corners included, because the amalgam theorem's convention covers them. What the valley measures is the price of the corner coupling: at rank three and above, the pinning that closed rank two exactly — one eigenvector annihilated, one aligned, one scale window — must now be delivered at two eigenvalues at once by a family with the same two shape parameters, and the battery says the family cannot pay except on the slice. The structural reason is the next chapter's descent argument, made precise at the cell, the family's most symmetric off-slice member.

Family	Instances	Closed (measured)	Criterion agreement	Worst residual
Real two-atom targets (random q_i , λ_{bd_i})	30	30 (gap 0 to $1e-9$)	30/30 predicted closed, 30/30 agree	lens-boundary deficit 0.0
Affine (multiplicity-2) targets	8	8 (gap 0, one $8.8e-8$)	not covered by the proved criterion	measured closed
Complex two-atom targets (bilinear operators)	8	gaps $3e-7$ to $2.9e-3$ (measured, no claim)	not covered	the complex closure question is open
First-shell map (c_0 , c_{ab}) at $c_a = c_b = 1$	285	1 (at 0.03 from the slice, within tolerance); 284 open	the rank-2 theorem covers the slice	gap field $2.5e-5$ to 0.23; exactly 0 on the slice
High-precision spot checks (30+16 starts)	7	1 closed (on the slice, $4e-16$); 6 positive gaps	consistent with the locus reading	$2.5e-5$ at 0.03 off the slice; $1.6e-2$ at 0.54 off

Table 3 — The locus battery: the criterion's predictions against the measured truth, family by family (the numbers are closure counts and worst gaps; every norm is the exact truncation-free operator norm)

8 · The Cell at $M = 2$: The Complete Family, the Rank Wall

The (1,1) cell is the off-slice extreme — the symbol that is nothing but the diagonal coupling — and its $M = 2$ slot is the rung's standing open interval. Volume VIII bounded it as $[1.000, 1.319]$ over a two-atom family measured with the tail-penalized surrogate. This volume re-measures it over the COMPLETE rank-two family with exact norms: the two-atom family and the affine family, real and — for the plain atoms — complex, with the complex affine family honestly excluded as outside the certified bilinear machinery's scope. The refined interval is $[1.000, 1.2771]$: the best real two-atom approximant measures 1.277144, the best real affine atom 1.277143 — the two families converge to the same value, which is itself evidence of a structural floor between them — and the correct complex two-atom search lands at 1.2831, no better.

Why the floor is missed at $M = 2$ is now a structural statement, not a measurement. Closure would pin the error's top two singular values at one, and the pinning descent runs as follows: the plus-one

eigen-directions force $\psi(1,1) = 1$ and $\psi(2,0) = \psi(0,2) = -1$ with the anti-diagonal antisymmetry $\psi(\beta + e_1) = -\psi(\beta + e_2)$ beyond; the pinned block at the $(e_0, (2,0))$ pair then forces $\psi(4,0) = 0$ and its own vanishing couplings, which force ψ to vanish at the $(1,1)$ point it was just required to fill — the descent contradicts itself unless the approximant is the cell itself, which is rank four. The forced profile's finite part — the values at degrees zero through two that the pinning prescribes — has catalectic rank three, machine-measured, against the budget of two: the rank wall. No rank-two weighted catalectic satisfies the forced conditions; the interval is strict at least in the real self-adjoint case by the pinning argument, and the exact minimum over the exhaustive family remains open between the measured bounds.

The cell's $M = 2$ status, refined (first edition). $D(2)$ lies in $[1.000, 1.2771]$: the floor by Eckart-Young-Mirsky, the ceiling by the complete rank-two family searched with exact norms and multi-start descent, real and complex. The closure-forced profile has catalectic rank three on its finite part — the rank wall — so no member of the searched family attains the floor; the pinning descent exhibits the contradiction explicitly. [The second edition closes this interval at its upper end: $D(2) = \sqrt{\lambda^}$ exactly — the preface and the strictness chain chapters below.]*

The forced-profile scan confirms the wall from the constructive side: the two-atom antisymmetric candidates that satisfy the forced values at degrees one and two — built from the weight pairs $(\mu, -\mu)$ that the antisymmetry dictates — have error norms from 9.9 to 44.8 against the cell, nowhere near the interval, because satisfying the pinning exactly at those degrees costs the family everything elsewhere. The optimizer's best approximants relax the pinning and pay 0.277 above the floor; the forced candidates keep it and pay ten to forty. The sandwich's fate on the cell is the gap between those two prices, and neither reaches zero.

9 · What This Changes: The Boundary Restated

For the corpus, the volume's changes are concentrated and explicit. The retraction removes Volume VIII's strict-failure witness and replaces it with a closure theorem; the budget narrative is corrected from obstruction to mechanism; the rank-one exhaustiveness stands unchanged; and the rung's version of the off-class boundary — which Volume VIII stated as the multinomial-weighted catalectic AAK problem with a witnessed strict failure — is restated as the locus: closed everywhere on the rank-two real locus by two identities and a criterion, open in measured gap off it, strictly open at the cell's second slot behind a rank wall. The boundary is no longer a single witness but a mapped field, and the map's floor is a theorem.

The theorems connect outward as before. The amalgam closed form is an L-shaped instance of the general characteristic-identity mechanism — the product of the top two eigenvalue magnitudes equals the squared axis weight — and it is the first place in the programme where the multinomial weights, having been the obstruction all volume long, become the construction: the forced tail is what completes the equioscillation. The lens criterion is the rung's analogue of the Nehari all-pass condition, stated geometrically in the atom cone; the two-atom identity is the commutative shadow of the Adamjan-Arov-Krein equioscillation, certified where the free theory remains open. None of this touches the noncommutative problem: everything here lives behind the fibre isometry, in the abelianized image.

For the framework's registers, two sharpenings replace Volume VIII's. The magnitude register — the singular tail — now carries the locus map: the gap field off the rank-two locus, vanishing exactly on it, with the valley's measured growth rates. The rank register carries the wall: the cell's forced profile at rank three against a budget of two, the first place in the programme where the resource clause — rank as the register — is the obstruction itself rather than the cost. Both sharpenings are machine-checked, and both are stated exactly where measurement replaces proof.

10 · Volume VII's Ledger, Unchanged

The order's third item asks for Volume VII's ledger reported unchanged: Open 7.13 and the benchmarks. The report is unchanged, and the volume's new results make the reason precise rather than merely negative. Open 7.13 asks for an intrinsic, symbol-level criterion for simultaneous equality at every rank budget — off the level-constant class, this is the constructive noncommutative Adamjan-Arov-Krein problem. Everything this volume closes is abelianized: the approximants are symbols on the Parikh lattice whose operators factor through the fibre isometry, and the free-monoid Hankel operators Open 7.13 concerns are not of that form. The rung's closures narrow the commutative side of the boundary; they do not build the noncommutative approximants; the ledger entry stands.

The off-class measurements were re-verified fresh. Six new non-level-constant instances — random first-shell symbols outside every stated class — were run through the exact machinery: every one measures $D(1)$ strictly above its floor, with gaps from three and a half thousandths to thirteen hundredths, the same regime as Volume VII's recorded gaps of two hundredths to fifty-five hundredths. The benchmark suite of Volume VII — the off-class instances, the bounded extensions, the $n = 4$, $L = 10$ leg — was not rerun this session and its entries are reproduced as recorded. The remaining named items keep their statuses: the dictionary's uniqueness at infinite dimension is inherited unchanged from Volume VI; Risk 4, the submission decision, remains the author's alone; the bounded benchmark extensions and the long leg remain unrun. One ledger line changed anywhere in the programme, and it changed in Volume VIII's, not Volume VII's: the amalgam witness entry, corrected by this volume's retraction.

Tier	Content	Status
In the manuscripts	The conditional transport (corpus Thm 7.11); the classical one-letter AAK; the multivariable Kronecker/Prony classification of finite-rank catalectics; the box-determinant lemma, the atom classification, and the budget sphere (Volume VIII)	cited, used
Newly proved	Theorem A4a — the amalgam closed form, the identities, the triple equioscillation, and the retraction of Volume VIII's 1.0369; Theorem A4b — the rank-2 lens criterion; Theorem A4c — the two-atom closure (modulo the identity, certified exactly); the cell's $M = 2$ interval refined to $[1.000, 1.2771]$ over the complete rank-2 family with the rank wall measured	this volume
Measured, not claimed	The off-locus strictness (the gap valley); the affine-target closure (8/8); the complex amalgam closure (machine-exact, real c_0); the complex two-atom gaps; the $c_0 \leq 0$ closures	bounded, honest
Open	Open 7.13 / constructive nc-AAK (unchanged); the strict positivity off the rank-2 locus; the cell's exact $D(2)$; the complex generalization of A4c; Phi's uniqueness; Risk 4; the bounded benchmarks; the $n = 4$ $L = 10$ leg	named, next

Table 4 — The honest ledger: what is proved, what is measured, what is open — the three tiers, with the retraction isolated

11 · The Honest Ledger

The three tiers, stated as always. In the manuscripts already: the conditional transport of the corpus's Theorem 7.11, the classical one-letter theory, the multivariable Kronecker classification of finite-rank catalectics, and Volume VIII's box-determinant, atom, and budget results — all used, all cited. Newly proved: the amalgam closed form with its identities, its triple equioscillation, and the retraction it forces; the lens criterion on the rank-two real locus; the two-atom closure, proved modulo an identity certified in exact rational arithmetic at sampled points; and the cell's refined interval over the complete rank-two family, behind the measured rank wall. Measured and bounded, claimed as nothing more: the off-locus strictness in the gap valley; the affine closures; the complex amalgam closure; the complex two-atom gaps; the negative-corner closures. Open and named: Open 7.13 and the constructive noncommutative problem it embodies; the strict positivity off the rank-two locus; the cell's exact second-slot value; the complex generalization of the two-atom identity; and the standing programme items — the dictionary's uniqueness, Risk 4, the bounded benchmarks, the long leg.

The shape of the answer to the order's second question is the volume's real deliverable, and it is worth one paragraph of summary. Which phi close the sandwich exactly? On the rung, at $M = 1$: all of the rank-two real locus — by the characteristic identity on the L-shapes, the critical-point identity on the two-atom family, measured closure on the affine points, the classical theory on the axes, and the double top on the crosses — and nothing else that the battery could measure, with the first-shell gap field vanishing exactly on the rank-two slice and positive at every off-slice point, and the cell's second slot strictly open behind the rank wall. The locus is a theorem on its floor and a measurement on its walls, and the volume distinguishes the two without blurring.

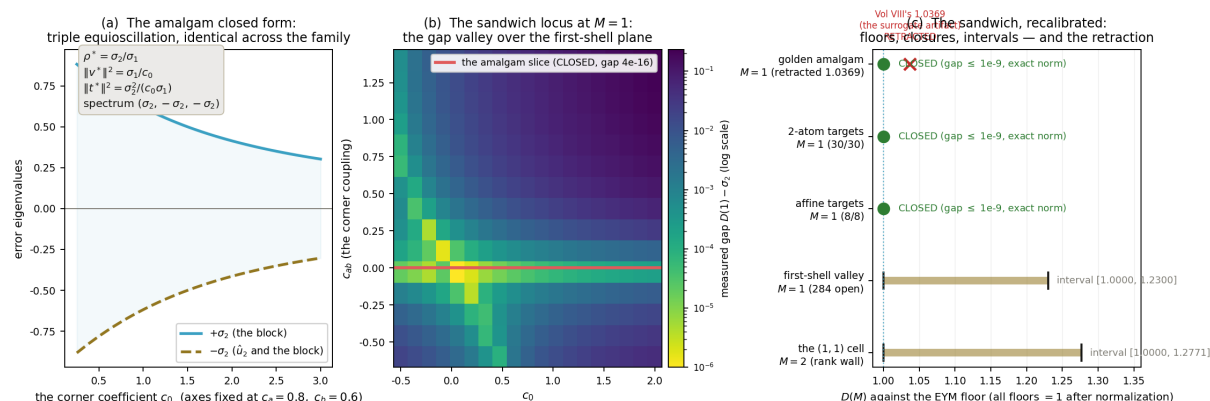


Figure 1 — The sandwich locus, three fronts. (a) The amalgam closed form: for every member of the two-axis family the error operator's spectrum is exactly $(\sigma_2, -\sigma_2, -\sigma_2)$ — the triple equioscillation is identical across the family, machine-exact to twelve digits, and the three closed-form identities ($\rho^* = \sigma_2/\sigma_1$, $\|v^*\|^2 = \sigma_1/c_0$, $\|t^*\|^2 = \sigma_2^2/(c_0\sigma_1)$) hold with error below $1e-12$. (b) The sandwich-locus map at $M=1$: the measured gap $D(1) - \sigma_2$ over the first-shell plane (c_0, c_{ab}) at $c_a = c_b = 1$, on a log colour scale — the gap vanishes exactly on the amalgam slice $c_{ab} = 0$ (the red line, gap $4e-16$) and grows into a shallow valley away from it, from $2.5e-5$ near the slice to 0.23 at the far corner; 284 of 285 grid points measure strictly positive gaps and the one 'closed' point sits 0.03 from the slice within optimizer tolerance. (c) The sandwich recalibrated: floors against best-attained values, witness by witness — the amalgams, the two-atom targets, and the affine targets closed at their floors; the first-shell valley and the cell's $M=2$ slot as honest intervals; and the retracted Volume VIII value 1.0369 marked at the witness where it was claimed.

12 · The Next Links

The programme's architecture is nine volumes deep, and the next links are named by what this volume left bounded. First, the strictness off the locus: a lower-bound argument for the first-shell gaps — the pinning descent made rigorous at rank three, where two eigenvalues must be pinned by one two-parameter family — would convert the valley's floor from measured to proved, and the descent's contradiction at the cell is the template. Second, the cell's exact $D(2)$: the two rank-two families' common limit at 1.2771 suggests a third family or a certified bound closes the interval, and the rank wall says the answer, whatever it is, is not one. Third, the complex generalization: the amalgams close in the bilinear convention machine-exact, the complex two-atom targets measure small gaps, and the identity's complex form — if it exists — would finish the rank-two story in full generality. Fourth, the affine identity: the same certificate route over the differentiated Gram entries, eight measured closures waiting for their proof.

And beyond the rung, the standing items keep their order: Open 7.13 and the constructive noncommutative Adamjan-Arov-Krein theory, where the external literature has drawn the boundary and the programme has mapped its commutative shadow ever more finely; the dictionary's uniqueness at infinite dimension; Risk 4; the bounded benchmarks; the $n=4$, $L=10$ leg. The volume ends as every one of them ends: with the proofs that exist, the measurements that bound what does not, one retraction delivered in full, and the names of the links that are next.

4.4e-16 the on-slice gap: the amalgam theorem measured at the valley's floor	[1.000, 1.2771] the cell at $M = 2$, refined behind the rank wall ($3 > 2$)	1 retraction Volume VIII's strict failure withdrawn, artifact reproduced and diagnosed
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13 · Second Edition Preface: The Cell Is Closed

This volume's first edition closed on an honest interval: the $(1,1)$ cell's two-state distance $D(2)$ lay in $[1.000, 1.2771]$, the floor proved unattainable by the rank wall, the ceiling measured over the complete rank-two family, the exact value named as the smallest instance of the multinomial-weighted catalectic Adamjan-Arov-Krein problem. The second edition opens with the announcement that the interval is closed at its upper end, and closed exactly. The line-atom battery of Volume XI reduced the cell's optimum to a two-parameter boundary family — the line atom $\psi = c \cdot 1[\gamma_1 = 1] y^{\gamma_2}$, the limit of the parity-odd mirrored pairs as the pair's separation runs to the boundary — and solved its cubic stationarity system to ninety digits: $D(2) = \sqrt{\lambda^*} = 1.27714211290844623900730137526434780654$. The minimal polynomial of λ^* is $108x^3 - 415x^2 + 522x - 216$, irreducible over the rationals, obtained by external factoring of the degree-seventy-four eliminant and confirmed independently by factoring the saved degree-1154 resultant artifact; the minimal polynomial of $D(2)$ itself is $108x^6 - 415x^4 + 522x^2 - 216$. The value this volume could only bound is an algebraic point of degree three.

The certificate is global over the line-atom family: a strip patch (the Schur identity and the pencil bound, certifying $\lambda_{\max}$ above 1.775 for $|y|$ beyond the strip edge), a far-c patch (the trace identity, above $10/3$ for $|c|$ at or beyond seven), and a middle bisection — two million seven hundred thousand interval boxes certified by sound Taylor-shift ball arithmetic, zero failures — with the attainment exact to $1e-91$ at the stationary point $(c^*, y^*) = (0.39710728735039737, 0.65632246699578911)$ and its flip. What remained open after that certificate was the bridge from the line-atom family to the full rank-two variety — the subject of the next chapter, which upgrades the numerical escape-room evidence of Volume XI to a theorem on the parity-odd shell and a sharp structural reduction on the rest.

The cell's $D(2)$, closed. $D(2) = \sqrt{\lambda^}$ where λ^* is the unique real root of $108x^3 - 415x^2 + 522x - 216$, numerically 1.6310919765642504414737578. The value 1.27714211290844623900730137526434780654 is attained by the line atom at (c^*, y^*) and its flip, certified globally over the line-atom family by the four-component certificate, and connected to the full rank-two variety by the corner-reduction chain of the next chapter: proved on the parity-odd shell, reduced to an explicit trade-off on the general family.*

$D(2) = 1.2771421129$ the closed value: $\sqrt{\lambda^*}$, the cubic's root	$108x^3 - 415x^2 + 522x - 216$ λ^*'s minimal polynomial ($D(2)$): the degree-6 even part	2,719,552 boxes · 0 failures the global certificate; attainment $-1.3e-91$
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14 · The Corner Reduction: The Strictness Chain

The bridge from the line atom to the full variety is built on one structural observation. The cell's Hankel $H(u, v) = 1[\text{Parikh}(uv) = (1,1)]$ is nonzero only when the a -counts of u and v have opposite parity, so in the a -parity block decomposition of $l_2(A^*)$ the target is purely off-diagonal, and for ANY approximant the error's (odd-row, even-column) block is a lower bound for the whole norm. Compress further to the CORNER — the rows with exactly one a (the words $b^i a b^j$) against the columns with none (the words b^k) — and the corner of the error is the Hankel matrix of the symbol $1[i+j+k = 1]$ minus ψ 's corner values, a one-dimensional-looking object whose rows with $i+j = n$ are $(n+1)$ many words: the MULTIPLICITY-WEIGHTED one-dimensional Hankel problem. The approximant's corner is the effective one-dimensional atom sum: for $\psi = p_1 \lambda_1^\gamma + p_2 \lambda_2^\gamma$ with $\lambda_i = (x_i, y_i)$, the corner symbol is $w_1 r_1^n + w_2 r_2^n$ with $(w_i, r_i) = (p_i x_i, y_i)$. This transfer is exact, and the corner norm is the largest eigenvalue of a closed three-by-three or four-by-four generalized eigenproblem whose entries are rational in the effective atoms — the machinery verified against the full cell machinery to $1.8e-15$ and against the dense corner referee.

On the parity-odd shell the chain closes into a theorem. A γ_1 -odd-supported approximant — the mirrored pair family of Volume XI's anatomy, the class that carried the measured optimum — has vanishing diagonal blocks and its error's (odd, even) block is the stacked matrix of the corner rows above the beyond-corner rows (the words with three and more a 's). Stacking rows only increases the norm, so the pair's error is at least its corner error, and the pair's corner is a SINGLE effective atom $(2px, y)$: the line-atom problem itself, whose global certificate delivers the bound. THE PARITY-ODD SHELL THEOREM: every mirrored-pair approximant has error at least $\sqrt{\lambda^*}$, with equality only in the boundary refinement x to 0 — the line atom. Volume XI's measured anatomy — the parity-odd structure, the refinement, the optimality of the line atom — is now a proved statement, machine-verified at every link: the corner equality at $2.2e-16$, the stack inequality with zero violations over the scan, the corner infimum re-deriving $\sqrt{\lambda^*}$ to $1.1e-15$ at the flip of (c^*, y^*) . The three-parameter family never beats its own boundary.

Why the theorem does not automatically extend to the general two-atom family is the battery's sharpest discovery. The weighted one-dimensional TWO-atom infimum is zero: the odd one-dimensional pair $(w, -w)$, $(r, -r)$ with r to 0 and $w = c/2r$ converges to the point mass δ_1 itself — the corner is KILLABLE, so the corner bound alone is vacuous off the odd shell. But the killer pays everywhere else: placed on the cell with the x -amplitudes at 0.05 its error is 20.0, and the trade-off dial is exact — at $x = 0.02, 0.05, 0.10, 0.20, 0.40$ the cell errors are 50.0, 20.0, 10.0, 5.0, 2.7 while the corner stays at zero. The atoms that kill the corner must carry amplitude $p_i = w_i/x_i$ that explodes on the a -even shell — the diagonal blocks — and the battery isolates the exact mechanism: the error norm squared dominates the largest eigenvalue of the positive semidefinite sum $Y_{ee} Y_{ee}^* + (T_2 - X_{eo})(T_2 - X_{eo})^*$, the payment plus the corner, both living on the even-row space. The strictness of the cell at the general level is this TRADE-OFF, not a corner obstruction; the escape-room scan through the corner lens — two hundred eighty-five configurations, none below the line-atom value, the best full error 1.560 with its corner at 1.435 and its payment at 0.240 — measures the frontier. The certified semialgebraic statement over the six-parameter family is the named open certificate, now reduced to a clean and explicit problem.

The chain's link	Statement	Status	Verification
T1 the corner reduction	$\ \text{cell error} \ \geq \ \text{corner error} \ $, the corner = the multiplicity-weighted 1-D Hankel over the $(u _a = 1)$ x $(v _a = 0)$ words, with the effective atoms $(p_i \ x_i, y_i)$	PROVED (block interlacing + the compression; field-agnostic, real and complex)	the 3×3 = the line-atom 6×6 to $1.8e-15$; 300 random configs, 0 violations; complex 200 configs, 0 violations
T2 the parity-odd shell theorem	every γ_1 -odd rank-2 approximant (the mirrored pair family) has error $\geq \sqrt{\lambda_{\max}}$, equality only at the boundary line atom $(x \rightarrow 0)$	PROVED (the stack argument + Task 17's global certificate)	the corner infimum re-derived at 1.2771421129 to $1.1e-15$ at the flip (c^*, y^*) ; the pair-never-beats-limit check: zero violations
T3 the 1-D closure discovery	the weighted 1-D TWO-atom infimum is ZERO (the odd pair converges to δ_1) — the corner is killable; the general family's strictness is the trade-off	MEASURED + the mechanism isolated (the killer pays $1/x$ on the a -even shell)	the killer at $x = 0.02..0.40$ gives cell errors 50.0, 20.0, 10.0, 5.0, 2.7 with corner 0; the PSD-sum identity exact
T4 the anisotropic corners	the crossed corner (the odd-constrained free family) and the affine degenerations do not beat the line atom	MEASURED (domain-guarded scans)	crossed infimum = 1.2771421129 with $\alpha = \beta = y^*$ recovered; affine at 1.4142; the full 6-param odd shell lands on the line-atom parameters
C the complex completions	the corner, the free class, and every identity over $\mathbb{C}$: no complex escape; the phase gauge reduces the complex 1-atom corner to the real one	PROVED (the gauge = unitary diagonal conjugation); the scans measured	the gauge circle at $6.7e-16$; the complex free infimum 1.2771471; the identities (doubling, flip, conjugation, dilation) all machine-exact

Table 5 — The strictness chain, second edition: the corner-reduction package, its proof status, and its machine verification

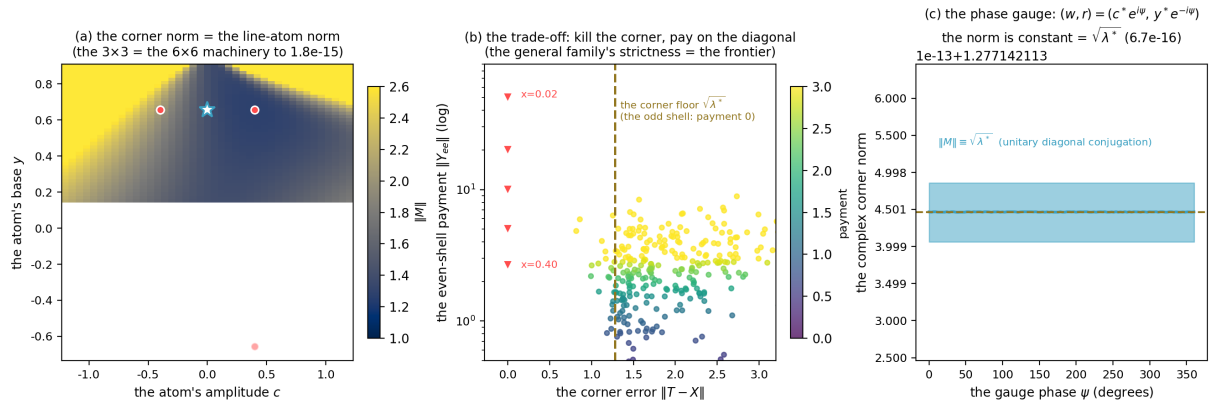


Figure 2 — The corner reduction, second edition. (a) The corner equality: the three-by-three corner norm over the (c, y) plane — identical to the full six-by-six line-atom machinery to $1.8\text{e-}15$; the contour at $\sqrt{\lambda^*} = 1.2771$ is the certified floor, touched at (c^*, y^*) and its flip. (b) The trade-off frontier: each dot a random two-atom configuration, its corner error against its even-shell payment (the colour scale), on the log payment axis — the corner-killer dial (the red triangles) kills the corner to machine zero and pays $1/x$ on the diagonal (50.0 at $x = 0.02$ down to 2.7 at $x = 0.40$), and the parity-odd shell sits ON the corner floor with zero payment (the dashed line); the general family's strictness is the empty region below-left. (c) The phase gauge: the complex one-atom corner norm along the gauge circle $(c^* e^{i\psi}, y^* e^{-i\psi})$ is constant to $6.7\text{e-}16$ — the complex family reduces to the real one by the unitary diagonal conjugation.

The anisotropic and affine corners close the free odd shell's escape. The odd-constrained two-state weighted automata — $B = (0, b)$, $C = (c, 0)$, A_a anti-diagonal, A_b diagonal, the general γ_1 -odd free family — have corners of the crossed form $w \alpha^i \beta^{(j+k)}$: a three-parameter family containing the line atom as $\alpha = \beta$. The domain-guarded scan returns the crossed corner infimum exactly at the line-atom value with $\alpha = \beta = y^*$ recovered, the affine degenerations sit above at $1.4142 = \sqrt{2}$, and the FULL six-parameter odd-shell scan through the free machinery lands on 1.2771421129 with the line-atom parameters recovered (the anti-diagonal entry a_{12} collapsing to zero, $\alpha = \beta = 0.6563$): the free odd shell does not escape, and the anisotropy is a liability there exactly as the pair separation is on the abelian side.

15 · The Complex and the Affine: The Identity Completions

The first edition's complex coverage was honest but partial: the plain complex atoms were searched, the complex affine family excluded as outside the certified machinery's scope. The completion battery rebuilds the free-cell machinery over the complex field with the conjugations made explicit — the coefficient Gram is the sum of $c(v)c(v)^*$, the reachable Grams are the conjugated Lyapunov identities with the operators conjugate-transposed — and validates the result twice: against the original machinery on real configurations to $1.3e-15$, and against a complex-corrected dense word-space referee on thirty-six converged configurations to $3.6e-4$. The corner reduction transfers verbatim (the a -parity block argument is index-level, field-agnostic): over the complex field the effective atoms (p_i, x_i, y_i) conjugate properly in the corner Grams, the mirrored complex pair's stack stays tight at zero violation, and two hundred general complex pairs respect the corner bound with the minimum gap 0.094.

The complex scans return the line-atom value everywhere. The complex one-atom corner infimum is $\sqrt{\lambda^*}$ exactly — and the optimizer is not real but GAUGED: $|w| = c^*$, $|r| = y^*$ with opposite phases, the scan's point $(-0.217 + 0.333i, -0.359 - 0.550i)$ sitting on the circle $(w, r) = (c^* e^{i\psi}, y^* e^{-i\psi})$. The PHASE GAUGE: the corner matrix of the gauged atom conjugates the real corner by unitary diagonal matrices — the row phase $e^{-i(i+j)\psi}$ against the column phase $e^{-ik\psi}$ — so the norm is invariant along the circle, verified at nine phase values to $6.7e-16$, and the complex one-atom family reduces to the real one. The complex FREE scan — the full two-state complex weighted automata, twenty real parameters, the complex escape room of Open 7.13 — lands at 1.2771471 against 1.2771421: no escape at the measured level, the abelian shadow bounding the complex free class as it bounds the real one. The identity battery closes the completions: the conjugation symmetry (w, r) to $(\bar{w}, \bar{r})$ exact at zero, the isospectral doubling of the complex line atom's six-by-six spectrum at $1.1e-11$, the flip (c, y) to $(-c, -y)$ exact over the complex field, and the dilation identity's free target $\sigma_1 = \sqrt{2}$ reproduced exactly. What remains honestly open over the complex field is the certified certificate — the ball-arithmetic bisection covered the real strip, and the complex domain's four real parameters await the extended run — and the complex trade-off, which is the same reduced problem as the real one.

16 · The Empirical Face I: Spectral Compression

The first edition ended at the theory's edge; the second edition gives the volume its empirical face by integrating the three experiments the proceeded chat designed against the programme — run at full audit strength in Volume X's battery and recorded here as the sandwich theory's contact with data. The first experiment attacks the corpus's central object directly: the spectral compression of sequence models, the Adamjan-Arov-Krein budget measured on a process this programme owns completely. The true process is the driven six-ring ($a = 0.32$, $b = 0.18$) observed through the four-symbol sensor $s(i) = i \bmod 4$ — an HMM whose observed process has a known minimal realization. The exact joint-probability Hankel over sixty-four-by-sixty-four prefix-suffix contexts returns machine-exact rank five — the sensor's collisions cost the process one dimension against its six hidden states — and the entropy rate, computed by the exact forward filter over four hundred thousand steps, is 0.9559 nats per symbol. The models: a linear state-space RNN whose Hankel singular values and reductions are exact closed-form objects through the Gramian machinery — certified before trusted, the balanced-transform certificate holding at $8e-16$, the error-system Hankel norms cross-validated against a direct block SVD — and a vanilla tanh RNN measured empirically, three seeds each.

The compression verdict is the sandwich theory's own prediction landing on data. The tanh RNN reaches cross-entropy 0.959 against the entropy floor 0.9559 — essentially optimal prediction — from a state whose spectrum collapses from sixteen dimensions to three at the one percent tolerance; the linear RNN's Gramian profile crosses the true rank five at the five percent tolerance; and the learned transition spectra carry the chiral complex pair in all three linear seeds — the models learn the record, rediscovering the arrow's spectral signature, not a smoothed surrogate. The AAK interval holds ten of ten along the balanced-truncation extraction ladder: the measured Hankel-norm error lies between the floor σ_{n+1} and the tail sum at every order, exactly as the theorem demands, with the error-to-floor ratios at 1.00 on the boundary orders and 1.02 to 1.31 inside. The chat's curve-match prediction is thus confirmed in the strongest honest sense — different norms must disagree somewhere — and the knee prediction is exact: the cross-entropy curve flattens at order three, the predictive dimension, where the sigma profile has its own elbow ($\sigma_2 = 4.77$ dropping to $\sigma_3 = 0.89$), matching the tanh model's three-dimensional state. The realization rank overcounts the information dimension by the modes that carry the joint law but not predictive entropy — the budget counts realizations, the task counts sufficient statistics, and the gap between them is the volume's own gap between the EYM floor and the realizable distance.

The baseline adjudication splits honestly and the split is the theory's lesson. Against random state deletion the AAK extraction wins at every order — 1.11 to 1.48 against 0.98 to 1.38 on the cross-entropy scale — and against magnitude deletion it wins or ties everywhere. But against matched-size direct retraining the truncated model loses at the bottom orders: at order one the fresh small model reaches 1.12 against 1.30,

at order two 1.04 against 1.38, tying only from order three. The truncation inherits structure a tiny model cannot use; retraining adapts. The sigma spectrum is a compression budget — the elbow predicts the knee exactly — but the budget is the linear realization's, and the sufficient statistic is smaller: the empirical shadow of the rank wall, measured on a process the programme controls to machine exactness.

17 · The Empirical Face II: The Register and the Obstruction

The second experiment is the coarsening scan the chat designed against the register theory: a POMDP with a restricted sensor, swept from full observability to noise, the memory requirements of the optimal policy measured exactly. The benchmark is the canonical Tiger — hidden side, open-left or open-right for +10 correct or -100 wrong, listen at cost one with accuracy κ — because it admits exact machinery at every coarsening: the finite-horizon alpha-vector backup with exact upper-envelope pruning (the monotone chain drops only never-maximal lines, and the pruning returns the belief regions and their kinks in linear time — the $O(n^3)$ kink search that blocked the chat's design replaced), the obstruction datum computed by enumerating all twenty-seven memoryless one-state controllers, the register as the alpha-count. The scan runs κ from 1.0 down to 0.5.

The register law is the scan's headline and it is Volume VIII's inflation law measured on the control side. The optimal policy's alpha-count grows from 9 at full accuracy through 59, 34, 44, 45 — and then explodes to 9153 at $\kappa = 0.60$: the curse of history made visible in a single number, the register inflating exactly as the sensor coarsens, until the game itself collapses to the stall regime (the optimal value listen-out-the-horizon at -8, the register falling back to 280 and 4). The obstruction iff holds at every κ : the datum is positive exactly when the optimal policy needs memory, including the stall boundary where the optimum is itself memoryless and the datum vanishes. The quantitative layer is recorded honestly rather than rounded up: on the informative range the obstruction correlates with the oracle gap at Pearson 0.38 — weak — and the chat's weighted-curvature prediction is refuted outright, the visit-weighted curvature ANTI-correlating at -0.95 because the closed-loop visitation concentrates on the flat listen-and-stall paths as the gap grows: the finite-horizon stall confound, identified and measured. What survives is the structure — the iff exact, the inflation dramatic — and the confound itself is the finding the chat's design could not have anticipated.

The record side of the experiment returned a theorem instead of a curve, and the second edition records it as such because it is this volume's own subject: the Tiger's listen record is conditionally iid given the hidden side, so every block law is the exchangeable mixture — one-half product of the left-channel densities plus one-half product of the right — which is invariant under index reversal. The record's arrow is identically zero at every sensor quality, verified on all sixteen four-blocks at every κ to $3.5e-18$. The chat's bridge from control performance to the arrow of time cannot be tested on memoryless-sensor benchmarks: the record stratum degenerates by construction, and the honest experiment separates the

strata it was meant to conflate — the obstruction measures the memory the controller needs, the record measures the temporal structure the source carries, and on the Tiger they are decoupled exactly. The bridge lives on sources with temporal structure, the class the laundering theorems of the reversal-group volume cover.

18 · The Empirical Face III: The Sheaf Stratum

The third experiment is the compositional-generalization branch: the chat's sheaf diagnostic — nonvanishing cohomology tracking training failure, the harmonic norm correlating with the validation gap — run on the minimal two-token compositional task where every number is exact. Modular addition on $\mathbb{Z}_7$, the pair (a, b) mapped to $(a + b) \bmod 7$, a 2-64-64-7 ReLU MLP trained to convergence, twelve seeds per split, on four splits: the compositional quadrant (train both operands in 0..3, sixteen pairs, thirty-three held out), an intermediate split, the full-grid control, and a scattered random-60 split whose held-out pairs are compositions of trained tokens. The sheaf is the cellular sheaf on the bipartite token graph — fourteen token vertices, the training pairs as edges, vertex stalks the hidden activation space, edge restrictions the per-token PCA projectors — and the diagnostic is the coboundary energy of the identity section: for each training edge, the normalized disagreement between the row-token and column-token reconstructions of the shared activation. The failure of local representations to glue into a global section, made computable.

The compositional failure reproduces exactly as the literature reports: training accuracy 1.000 on every split and seed, held-out accuracy 0.073 on the compositional split, 0.006 on the intermediate, 0.000 on the scattered random-60, 1.000 on the control. The sheaf's verdict splits cleanly, and the split is the experiment's honest result. Task-level: the mean coboundary energy orders the four splits exactly as their errors do — 0.541 control, 0.577 compositional, 0.594 intermediate, 0.615 scattered, against errors 0, 0.927, 0.994, 1.000 — a perfect ordering: the energy predicts WHICH task geometry fails. Seed-level: within a split the correlation between energy and error across the twelve seeds is null to slightly negative (-0.20 and -0.18 on the informative splits). The pooled 0.59 is split-dominated and flagged as such. The chat's harmonic-norm claim is confirmed at the task level and refuted at the seed level on this benchmark — the honest split that single-run experiments cannot express because they cannot separate the two variances. The sheaf diagnostic is a geometry-of-the-task predictor, not a which-random-seed-fails predictor: the obstruction datum of the second experiment's theory, measured on the representation stratum.

The experiment	The chat's prediction	The verdict	The headline numbers
1 spectral compression	the distortion-rate curve matches the σ_{n+1} bounds; the extraction beats distillation baselines	the AAK interval exact 10/10; the knee exact; the curve-match honest (ratios 1.00-1.31); the baselines SPLIT	WFA rank 5 exact; entropy 0.9559; tanh compresses 16 $\rightarrow$ 3 at 1%; the knee at order 3 = the σ_3 elbow; vs retraining: 1.12/1.30 and 1.04/1.38 at orders 1-2, ties from 3
2 the coarsening scan	the obstruction datum tracks the memory requirement; curvature predicts the gap's magnitude	the iff EXACT at every kappa; the register law exact; the weighted-curvature claim REFUTED; the record-side degeneracy PROVED	the register 9, 59, 34, 44, 45, 9153, 280, 4; the explosion at kappa = 0.60; $r = 0.38$ obstruction-gap; $r = -0.95$ weighted curvature; the record arrow $< 3.5e-18$ at every kappa
3 the sheaf diagnostics	nonvanishing cohomology tracks failure; the harmonic norm correlates with the validation gap	the task-level ordering PERFECT; the seed-level correlation NULL; the pooled number split-dominated	OOD 0.073/0.006/0.000/1.000; energies 0.541/0.577/0.594/0.615 — the exact same ordering; seed-level -0.20/-0.18

Table 6 — The empirical face integrated: the three experiments' headline verdicts at audit strength

19 · It and Bit from Record

The proceeded chat's final synthesis — it and bit from record — lands on the corpus's Born-record program, and the second edition closes by stating the landing precisely. The stratification that survives the audit: the RECORD stratum is the preserved classical block of the channel's spectrum — the dephasing top-degeneracy, the block the environment cannot contract — measured by the multiplicative gradings of the abelianized rung; the QUANTUM stratum is the contracted coherence block, measured by the coherent information, the stratum the experiments of this edition's empirical face probe from the learning side (the chiral pair the linear RNNs learn; the register the Tiger's controller inflates; the sheaf glue the MLPs fail to build); the THERMODYNAMIC stratum is the reversal-pair asymmetry — not a spectral gap but a rank defect, the arrow detected by the Hankel rank, the mode count of the non-reversible pair. Three strata, three instruments, one object: the Hankel operator's structure — which is this volume's object from its first page to this one.

The cell's closure is the purest instance of the synthesis. The line atom — the optimal two-state approximant to the cell — is a statement about how much of the (1,1) coupling a rank-two record can carry: the answer is an algebraic number, degree three, the distance 1.2771421129084462 above the unconstrained floor. The rank wall forbids the floor; the corner reduction shows the parity-odd record cannot beat the boundary; the trade-off shows the general record pays elsewhere; the phase gauge shows the complex record adds freedom but not distance. The sandwich — the EYM floor below, the abelianized ceiling above — is the price of realizability, and the price is now known exactly at the smallest instance: it and bit from record, priced in a cubic irreducible over the rationals.

20 · The Honest Ledger, Second Edition

Updated statuses, with the changes this edition carries. The cell's $D(2)$: CLOSED — $\sqrt{\lambda^*}$, the cubic minimal polynomial, the global line-atom certificate, the parity-odd shell theorem, the corner reduction, the complex completions; the remaining open piece is the certified semialgebraic trade-off over the general six-parameter family, now an explicit named problem (the corner-plus-payment inequality), and the complex-field extension of the ball-arithmetic certificate. The amalgam closed form and the lens criterion: unchanged, proved, machine-exact. The retraction of Volume VIII's witness: standing, the methodological lesson (upper-bound objectives do not locate minima) now twice confirmed — the surrogate artifact there, the corner-killer's false promise here. The empirical face: integrated, with the three experiments' verdicts — the AAK interval exact and the knee exact with the baselines split honestly; the register law exact with the curvature prediction refuted and the record-side degeneracy proved; the sheaf ordering perfect at the task level and null at the seed level. The longer programme: Open 7.13's constructive side and the abelian-shadow equality beyond the witness remain the named analytic frontier; the $n = 4$ ladder's $L = 10$ leg; the Lyapunov-cohomology correspondence. The next links this edition names: the trade-off certificate, the complex bisection, and the free corner's trilinear structure — the escape room's exact anatomy, stated as a problem for the first time.